

Dataset Requirements

Dataset Overview

You should be able to select a dataset that:

- Demonstrates your ability to tackle complex problems.
- Enables creative and original approaches to analysis.
- Satisfies project requirements for supervised and unsupervised learning.

Where can I find potential datasets?

- Look for datasets that haven't been widely explored.
- Here is a list of some data sources that you can consider. Note that **not** every dataset in the sources is suitable for the project. So, you still need to check the information on each dataset carefully against the requirements below.
 - Check out [the dataset ideas list](#) from the Capstone course. You can use the same dataset in Milestone 2 and Capstone.
 - Here is a U-M datasets Airtable ([here](#)) curated by UMSI.
 - U.S. Government data sources (e.g., [data.gov](#))
 - Please note that the [UCI Machine Learning Repository](#) datasets can be a good starting point, but by themselves are so commonly used (and/or limited in size) that they must be combined with additional datasets to be approved for project use.
 - University of Michigan, Institute of Social Research's [ICPSR's datasets](#)

General Requirements

Here is a list of requirements you must follow for data selection for course projects.

1. Ensure the dataset and methods **do not** have readily available solutions or tutorials.
 - Using datasets with existing solutions or tutorials (e.g., Kaggle competitions) will not allow you to

Using datasets with existing solutions or tutorials (e.g., Kaggle competitions, GitHub solutions) is fine, but you must demonstrate your original thinking, analytical skills, or creativity.

- Example: Avoid datasets like the "Titanic Survival Prediction" or "MNIST Digits Classification," as they have numerous tutorials and complete solutions readily available online.

2. If you are interested in widely used datasets, then create a new dataset by combining or transforming existing datasets in a new way.

- If you are interested in widely used datasets, then manipulating them to create a new problem, such as by joining with a significant second dataset, can help you demonstrate creativity.
- Example: If the COVID-19 case dataset has been explored extensively for time-series predictions, consider combining it with economic data (e.g., unemployment rates) to predict economic trends during the pandemic.
- If you take this approach, discuss your ideas with instructors early. Present a clear plan for how you will transform or merge datasets to define a new research question.

3. Use widely explored datasets in innovative ways.

- Even if a dataset has been extensively analyzed, you can still add value by applying new methods or asking new questions.
- Example: The Census Income Dataset (also known as the "Adult" dataset) is widely used for binary classification (e.g., predicting if an individual earns more than \$50K). Instead of replicating traditional classification tasks, you could approach the dataset from a new perspective, such as using unsupervised learning to identify clusters of individuals based on demographic and work-related features (e.g., education, occupation, hours worked) or investigating fairness and bias in machine learning by analyzing how model performance differs across demographic groups (e.g., gender, race, or

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able to clearly explain how your analysis will differ from what has already been done with the dataset.

4. Ensure the dataset is freely accessible to instructors

- Instructors need to verify your methods, results, and implementation. Using datasets with restrictions prevents thorough evaluation.
- Avoid datasets requiring NDAs, proprietary tools, or paid access. If your dataset is sensitive, anonymize features (e.g., renaming variables without revealing their meaning).
- Share public links to datasets. If the data isn't public, consult instructors to confirm its use. In general, it may be fine to use a non-public dataset, as long as the instructors have access.
- If the dataset is huge, the instructors should at least have access to a representative sample and be able to verify the operation of your code on that sample.

5. Select a dataset that offers sufficient challenge: the prediction problem should not be too easy. It should also not be extremely difficult. Ask yourself: could humans have consistent above-random success on this prediction task? If the answer is 'no' then you probably want to reconsider the problem.)
 - Projects should showcase your technical skills by addressing challenges such as data volume, variable complexity, or incomplete information.
 - Example: A dataset with 100 rows is insufficient for robust supervised learning. An e-commerce dataset with thousands of customer transactions and product attributes might be better.
 - Projects in the genre of *stock price prediction* are specifically discouraged unless you can show a novel component or problem formulation that goes beyond using standard historical data to predict standard asset values. For example, predicting stock price *volatility* is actually a tractable problem.
 - Assess the dataset for the following:
 1. Supervised Learning: Ensure it includes labeled outcomes. You will compare and evaluate different learning frameworks and feature representations.
 2. Unsupervised Learning: Check the dataset has enough meaningful structure for clustering or dimensionality reduction.
 6. Consult the instructors before finalizing your dataset.
 - Datasets that initially seem appropriate may not meet the project requirements. Early feedback will save time and ensure approval.
 - Assuming your dataset meets requirements without confirmation can lead to rejection.
-  Share with your advising instructor (coach) detailed information about your proposed dataset, including:
1. A link to the dataset or repository.
 2. A brief dataset description (including at a minimum: format, size, variables, data source).

Building on a Milestone I (or Previous Course) Project

If you plan to continue your Milestone I project (or any previous course project) for Milestone II, your project must satisfy all dataset selection conditions outlined above.

1. Provide a clear breakdown of existing vs. new work. Specifically, in your project proposal, explicitly differentiate between:
 - Existing code: Code reused from your previous project, such as data manipulation, cleaning, or integration.

- New code: Code you plan to develop for Milestone II, which should include entirely new methods, analyses, and implementations.
2. Restrictions on code reuse:
- You **cannot** reuse any supervised or unsupervised learning code for which you received credit in Milestone I (or another course).
 - Reused code should be limited to data preprocessing or preparation steps.
 - Example: If your Milestone I project involved building a classification model using logistic regression, you can reuse the data preprocessing steps (e.g., handling missing values, feature encoding). Still, you must implement new methods (e.g., decision trees, ensemble learning) for Milestone II.
3. Disclosure requirement: Failure to clearly disclose reused code or previously conducted work may be considered academic misconduct. Ensure transparency in your proposal to avoid issues.

Using Social Media or Web Data for Course Projects

Social media data collection needs a separate discussion because, while social media platforms often appear to offer publicly accessible data, their terms of service and data usage policies are complex and restrictive. The ease of access and the appeal of analyzing large, user-generated datasets can attract students. However, failing to navigate the ethical, legal, and privacy considerations properly may lead to unintended violations. Addressing this topic separately ensures students know these nuances and make informed, compliant decisions in their projects.

What You Should Do

1. Use official APIs provided by social media/web platforms.
 - Official APIs ensure legal and ethical data access, even though they may come with restrictions (e.g., rate limits or fees).
 - Example:
 - X (Twitter)'s API allows you to collect tweets with specific keywords but may limit the number of tweets per request.
 - Reddit provides an API for accessing posts and comments without violating their policies.
2. Check the Terms of Service (ToS) and Policies of Platforms before collecting data.
 - Social media platforms often explicitly prohibit web scraping in their ToS, and violating these terms can pose legal risks.
 - Example:
 - Indeed.com prohibits all scraping, as stated in its [privacy FAQ](#). Even if the data on Indeed.com seems

publicly accessible, scraping it is a violation.

- Amazon.com similarly restricts scraping but provides resources like publicly released Amazon review datasets that comply with its policies.

3. Explore open datasets with clearly stated licenses.

- Open datasets often have licenses (e.g., Creative Commons or MIT) that allow legal usage.
- Example:
 1. Use open-source datasets like the [Amazon Review Dataset](#), which aggregates reviews into a structured format over time.
 2. Platforms like [data.world](#), [Zenodo](#), and research lab repositories often host social media datasets under open licenses.

4. Consult with the instructors for data acquisition processes.

- If your data needs are specific or not publicly available, you may need to involve the university's legal team to negotiate access. This might not be preferable as completing the required negotiation procedure will take a long time.
- Example: If you plan to analyze, for example, LinkedIn posts or Facebook group activity, the best approach might be directly contacting the company or exploring publicly available datasets compiled by others with clearly stated licenses.

5. Prioritize ethical and transparent use of data.

- Ethical considerations around user privacy and data ownership are critical in social media research.
- Example: Avoid datasets that include personally identifiable information (e.g., usernames, email addresses) unless the data is anonymized or public with explicit permission.

6. Use open-source tools and platforms compliant with policies.

- Many research tools exist that comply with social media platforms' policies for data collection.
- Example: Use tools listed in the X's formal webpage ([here](#)) for Twitter data collection and PRAW ([here](#)) for Reddit data collection through APIs instead of unauthorized scraping methods.

What You *Shouldn't* Do

1. Don't use web scraping tools without verifying platform policies.

- Scraping websites that explicitly prohibit it can result in legal issues, blocked IPs, or ethical violations.
- Example: Even if blogs or Reddit posts provide code for scraping Indeed.com, its Terms of Service (ToS) prohibit scraping, making such actions inappropriate.

2. Don't assume public websites are automatically legal to scrape.

- Public accessibility does not equal permission. Most platforms have terms that regulate automated access.
- Example: Just because Amazon product reviews are viewable on their website doesn't mean scraping is allowed. Instead, use authorized open datasets like the [Amazon Review Dataset](#).

3. Don't use datasets without clear licensing or usage terms.

- Ambiguous licensing can lead to compliance and legal issues. Confirm before use if the license (e.g., CC-BY, MIT) isn't specified.
- Example: Avoid datasets from unverified GitHub repositories or platforms that don't state data ownership and permissions explicitly.

4. Don't ignore privacy concerns with user-generated Content.

- User-generated data (e.g., reviews and posts) often has implicit privacy concerns, even if publicly visible.
- Example: Collecting user reviews from Amazon.com, or social media comments without anonymization could breach user privacy expectations.

5. Don't rely on unverified claims about data legality.

- Blogs, forums, or community posts that claim data is legal to scrape often lack credible references. Relying on these can expose your team and the university/school to risks.
- Example: A blog stating that scraping public Facebook pages is legal does not override Facebook's ToS, which prohibits automated scraping.

6. Don't proceed without instructor guidance if unsure about data compliance.

- If you have doubts about whether data collection methods or datasets meet legal or ethical guidelines, instructors can provide guidance or escalate the issue.
- Example: If you want to analyze LinkedIn job postings, consult the instructor to explore legitimate approaches instead of scraping without approval.

Remember: Always consult with your instructors early and often to ensure your dataset aligns with project expectations.